

Overview of antibacterial, antitoxin, antiviral, and antifungal activities of tea flavonoids and teas.



Tea leaves produce organic compounds that may be involved in the defense of the plants against invading pathogens including insects, bacteria, fungi, and viruses. These metabolites include polyphenolic compounds, the six so-called catechins, and the methyl-xanthine alkaloids caffeine, theobromine, and theophylline. Postharvest inactivation of phenol oxidases in green tea leaves prevents oxidation of the catechins, whereas postharvest enzyme-catalyzed oxidation (fermentation) of catechins in tea leaves results in the formation of four theaflavins as well as polymeric thearubigins. These substances impart the black color to black teas. Black and partly fermented oolong teas contain both classes of phenolic compounds. A need exists to develop a better understanding of the roles of polyphenolic tea compounds in food and medical microbiology. This overview surveys and interprets our present knowledge of activities of tea flavonoids and teas against foodborne and other pathogenic bacteria, virulent protein toxins produced by some of the bacteria, virulent bacteriophages, pathogenic viruses and fungi. Also covered are synergistic, mechanistic, and bioavailability aspects of the antimicrobial effects. Further research is suggested for each of these categories. The herein described findings are not only of fundamental interest, but also have practical implications for nutrition, food safety, and animal and human health.